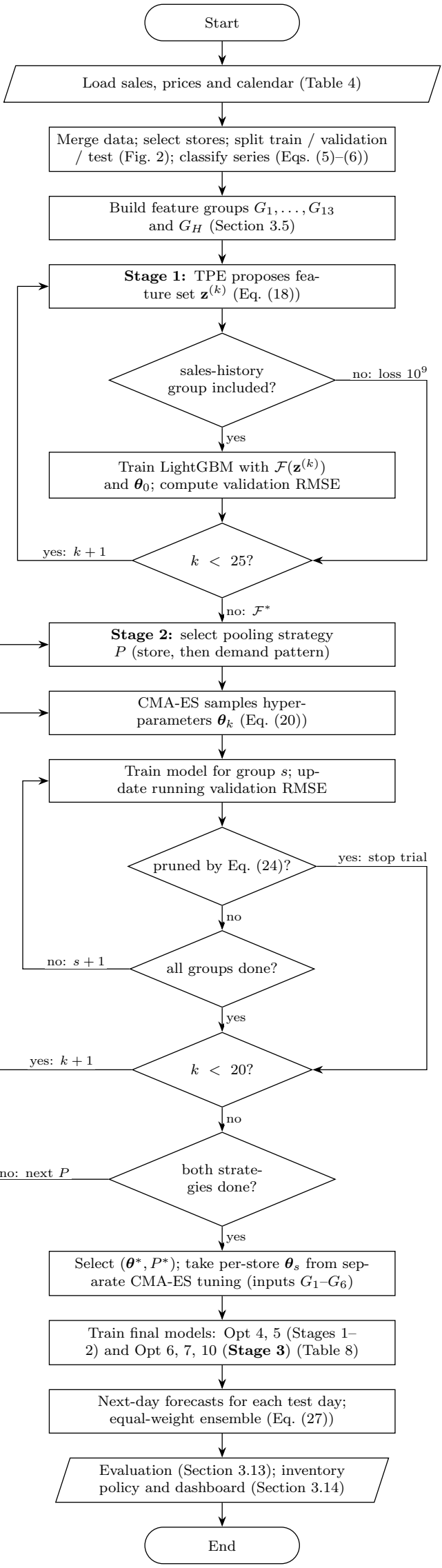

Start
Load sales, prices and calendar (Table 4)
Merge data; select stores; split train / validation / test (Fig. 2); classify series (Eqs. (5)–(6))
Build feature groups $G_1, \ldots, G_{13}$ and $G_H$ (Section 3.5)
**Stage 1:** TPE proposes feature set $\mathbf{z}^{(k)}$ (Eq. (18))
sales-history group included?
no: loss $10^9$
yes
Train LightGBM with $\mathcal{F}(\mathbf{z}^{(k)})$ and $\boldsymbol{\theta}_0$; compute validation RMSE
yes: $k+1$
$k < 25$?
no: $\mathcal{F}^*$
**Stage 2:** select pooling strategy $P$ (store, then demand pattern)
CMA-ES samples hyperparameters $\boldsymbol{\theta}_k$ (Eq. (20))
Train model for group $s$; update running validation RMSE
pruned by Eq. (24)?
yes: stop trial
no
no: $s+1$
all groups done?
yes
yes: $k+1$
$k < 20$?
no
no: next $P$
both strategies done?
yes
Select $(\boldsymbol{\theta}^*, P^*)$; take per-store $\boldsymbol{\theta}_s$ from separate CMA-ES tuning (inputs $G_1$–$G_6$)
Train final models: Opt 4, 5 (Stages 1–2) and Opt 6, 7, 10 (**Stage 3**) (Table 8)
Next-day forecasts for each test day; equal-weight ensemble (Eq. (27))
Evaluation (Section 3.13); inventory policy and dashboard (Section 3.14)
End